

API Reference

This document is the API reference for the agent marketplace backend.

The implementation is HTTP-first, versioned under /v1 for domain routes, and uses FastAPI-style JSON bodies and status codes.

Common Mechanics

Base URLs

- Root and health routes: /, /health, /health/live, /health/ready
- Domain routes: /v1/...

Common Headers

- Authorization: Bearer <jwt-or-api-key> for protected routes that accept generic bearer auth
- Idempotency-Key for invoke requests and provider payout requests
- X-Request-ID for request correlation
- PAYMENT-REQUIRED on 402 Payment Required responses
- PAYMENT-SIGNATURE on paid retry requests produced by the x402 client
- PAYMENT-RESPONSE on successful paid invoke responses
- The service uses X-Request-ID for request correlation and echoes it when present

Common Error Shape

Most route handlers raise FastAPI HTTP errors with JSON bodies shaped like:

```
{
  "detail": "human-readable message"
}
```

Authentication Matrix

- Public: root and health routes, discovery routes, quote creation, and auth nonce / verify / refresh
- Generic bearer auth: provider authoring, invoke, invocation reads, provider earnings, ledger, public payout reads, and admin routes
- JWT-only: API-key lifecycle, account self-service, wallet rotation, and provider payout execution

Health

GET /

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"name":"Agent Marketplace Backend","status":"ok","docs":"/docs","health":"/health","ready":"/health/ready"}
- Errors: none expected in the normal path

GET /health

- Auth: none
- Request: no body, no parameters

- Success: 200 with {"status":"ok"}
- Errors: none expected in the normal path

GET /health/live

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"status":"ok"}
- Errors: none expected in the normal path

GET /health/ready

- Auth: none
- Request: no body, no parameters
- Success: 200 with {"status":"ok"}
- Errors: 503 with {"detail":"..."} when readiness checks fail

Auth

GET /v1/auth/nonce

- Auth: none
- Request: query parameter address with a 42-character wallet address
- Success: 200 with {"nonce":"..."}
- Errors: 422 if the address is malformed

Example response:

```
{
  "nonce": "f1d0b88e3f8b4dce"
}
```

POST /v1/auth/verify

- Auth: none
- Request: body with message and signature
- Success: 200 with access_token, refresh_token, and the authenticated account
- Errors: 401 when the signature or message is invalid

Example request:

```
{
  "message": "127.0.0.1 wants you to sign in with your Ethereum account:\n0x...\n\nURI:
    http://127.0.0.1:8000\nVersion: 1\nChain ID: 84532\nNonce: f1d0b88e3f8b4dce\nIssued At: 2026-03-
    17T12:00:00Z",
  "signature": "0x..."
}
```

Example response:

```
{
  "access_token": "eyJ...",
  "refresh_token": "eyJ...",
  "account": {
    "id": 1,
```

```
"wallet_address": "0x...",
"account_type": "human",
"is_admin": false,
"display_name": "Demo Account",
"created_at": "2026-03-17T12:00:00Z",
"updated_at": "2026-03-17T12:00:00Z"
}
}
```

POST /v1/auth/refresh

- Auth: none
- Request: body with refresh_token
- Success: 200 with a fresh access_token
- Errors: 401 when the refresh token is invalid or expired

POST /v1/auth/api-keys

- Auth: JWT-only
- Request: body with optional name and expires_at
- Success: 201 with metadata plus the plaintext api_key once
- Errors: 422 when the request is invalid

GET /v1/auth/api-keys

- Auth: JWT-only
- Request: no body
- Success: 200 with a list of API-key metadata
- Errors: none expected in the normal path

DELETE /v1/auth/api-keys/{api_key_id}

- Auth: JWT-only
- Request: path parameter api_key_id
- Success: 204 with no body
- Errors: 404 when the key does not exist

Account

GET /v1/account/me

- Auth: JWT-only
- Request: no body
- Success: 200 with the current account
- Errors: 404 when the account cannot be found

PATCH /v1/account/me

- Auth: JWT-only
- Request: body with optional display_name
- Success: 200 with the updated account
- Errors: 422 when no updatable field is provided or the data is invalid

Example request:

```
{  
  "display_name": "Demo Provider"  
}
```

POST /v1/account/wallet

- Auth: JWT-only
- Request: body with `wallet_address`
- Success: 200 with a wallet-change nonce and expiry
- Errors: 409 when the requested change conflicts with account state, 422 for invalid input

POST /v1/account/wallet/confirm

- Auth: JWT-only
- Request: body with `message` and `signature`
- Success: 200 with fresh tokens and the updated account
- Errors: 409 when the wallet change flow is not valid or has expired

Provider Authoring

POST /v1/provider/services

- Auth: generic bearer
- Request: body with `slug`, `name`, `summary`, and optional `description`
- Success: 201 with the created service
- Errors: 409 on conflicts, 422 for invalid input

GET /v1/provider/services

- Auth: generic bearer
- Request: no body
- Success: 200 with the authenticated account's services
- Errors: none expected in the normal path

GET /v1/provider/services/{service_id}

- Auth: generic bearer
- Request: path parameter `service_id`
- Success: 200 with the owned service detail
- Errors: 404 when the service does not exist or is not owned

PATCH /v1/provider/services/{service_id}

- Auth: generic bearer
- Request: body with optional `name`, `summary`, and `description`
- Success: 200 with the updated service
- Errors: 409 on invalid state transitions, 422 for invalid data

POST /v1/provider/services/{service_id}/tags

- Auth: generic bearer
- Request: body with `tags` as the full replacement list

- Success: 200 with the service and updated tags
- Errors: 404 if the service is not owned, 422 for invalid input

POST /v1/provider/services/{service_id}/publish

- Auth: generic bearer
- Request: no body
- Success: 200 with the published service
- Errors: 409 when publish requirements are not met

POST /v1/provider/services/{service_id}/endpoints

- Auth: generic bearer
- Request: body with endpoint key, name, summary, description, access_mode, request_schema, response_schema, timeout_seconds, is_enabled, and optional pricing
- Success: 201 with the created endpoint
- Errors: 409 on invalid service state, 422 for schema or pricing problems

PATCH /v1/provider/endpoints/{endpoint_id}

- Auth: generic bearer
- Request: body with optional endpoint fields and optional pricing
- Success: 200 with the updated endpoint
- Errors: 404 if the endpoint is not owned, 409 on invalid state transitions, 422 for invalid input

PUT /v1/provider/endpoints/{endpoint_id}/upstream

- Auth: generic bearer
- Request: body with base_url, path, http_method, and optional config
- Validation: path must start with / and must not include a scheme, host, query string, or fragment
- Success: 204 with no body
- Errors: 404 if the endpoint is not owned, 409 on invalid state transitions, 422 for invalid input

Example upstream config payload:

```
{
  "base_url": "https://provider.example.com",
  "path": "/invoke",
  "http_method": "POST",
  "config": {
    "auth": {
      "type": "hmac_sha256",
      "key_id": "gateway-key",
      "secret": "gateway-secret"
    }
  }
}
```

Discovery

GET /v1/services

- Auth: none
- Request: no body
- Success: 200 with the public service list

- Errors: none expected in the normal path

GET /v1/services/{service_id_or_slug}

- Auth: none
- Request: path parameter `service_id_or_slug`
- Success: `200` with the public service detail
- Errors: `404` when the service does not exist or is not publicly available
- Lookup rule: a path segment made only of digits is treated as a service id, not a slug

GET /v1/services/{service_id_or_slug}/schema

- Auth: none
- Request: path parameter `service_id_or_slug`
- Success: `200` with the enabled endpoint schemas
- Errors: `404` when the service does not exist or is not publicly available

GET /v1/services/{service_id_or_slug}/pricing

- Auth: none
- Request: path parameter `service_id_or_slug`
- Success: `200` with the enabled endpoint pricing data
- Errors: `404` when the service does not exist or is not publicly available

POST /v1/services/{service_id_or_slug}/quote

- Auth: none
- Request: body with `endpoint_key` and `payload`
- Payload shape: `payload` may be any JSON value that matches the endpoint request schema, not only an object
- Success: `201` with a quote bound to the request payload
- Errors: `404` when the service or endpoint cannot be found, `409` when the quote cannot be created for the current state

Example request:

```
{
  "endpoint_key": "paid-summary",
  "payload": {
    "message": "Summarize this marketplace request."
  }
}
```

Example response:

```
{
  "id": 1,
  "service_id": 1,
  "endpoint_key": "paid-summary",
  "pricing_type": "fixed_per_call",
  "amount_minor": 100,
  "currency": "USD",
  "request_hash": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef",
  "service_revision_id": 1,
  "service_change_token": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef",
  "expires_at": "2026-03-17T12:05:00Z",
}
```

```
"created_at": "2026-03-17T12:00:00Z"
}
```

Invoke

POST /v1/invoke/{service_id_or_slug}

- Auth: generic bearer
- Request: body with endpoint_key, payload, and optional quote_id
- Payload shape: payload may be any JSON value that matches the endpoint request schema, not only an object
- Required header: Idempotency-Key
- Success: 200 with the invocation record, or 402 for unpaid paid-endpoint requests
- Errors: 404 for missing services or invocations, 409 for conflicts or in-progress requests, 502 for upstream failures, 504 for upstream timeouts
- Invocation status values: in_progress, succeeded, failed
- Upstream idempotency: the forwarded X-Agent-Marketplace-Invocation-Id header is stable across retries for the same invocation and should be treated as the provider-side idempotency key

Example free-invoke request:

```
{
  "endpoint_key": "free-ping",
  "payload": {
    "message": "hello from the free route"
  }
}
```

Example invocation response:

```
{
  "id": 1,
  "service_id": 1,
  "endpoint_key": "free-ping",
  "access_mode": "free",
  "quote_id": null,
  "idempotency_key": "consumer-example-1",
  "request_hash": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef",
  "status": "succeeded",
  "upstream_status_code": 200,
  "response_payload": {
    "message": "pong",
    "mode": "free"
  },
  "error_message": null,
  "created_at": "2026-03-17T12:00:00Z"
}
```

GET /v1/invocations/{invocation_id}

- Auth: generic bearer
- Request: path parameter invocation_id
- Success: 200 with the invocation detail
- Errors: 404 when the invocation is not accessible

GET /v1/invocations

- Auth: generic bearer
- Request: no body
- Success: 200 with the caller's invocation list
- Errors: none expected in the normal path

Finance

GET /v1/provider/earnings

- Auth: generic bearer
- Request: no body
- Success: 200 with earnings totals grouped by currency
- Errors: none expected in the normal path

GET /v1/provider/ledger

- Auth: generic bearer
- Request: no body
- Success: 200 with ledger entries
- Errors: none expected in the normal path

GET /v1/provider/payouts

- Auth: generic bearer
- Request: optional status query parameter
- Success: 200 with payout summaries and payout records
- Errors: none expected in the normal path

POST /v1/provider/payouts

- Auth: JWT-only
- Request: no body, but Idempotency-Key is required
- Success: 200 with payout-request results
- Errors: 409 when a duplicate or conflicting request is submitted

Example response:

```
{
  "idempotency_key": "example-provider-payout-1",
  "requested_count": 2,
  "sent_count": 2,
  "failed_count": 0,
  "payouts": []
}
```

Admin

POST /v1/admin/services/{service_id}/suspend

- Auth: admin actor
- Request: body with reason
- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

POST /v1/admin/services/{service_id}/restore

- Auth: admin actor
- Request: body with reason
- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

POST /v1/admin/services/{service_id}/delist

- Auth: admin actor
- Request: body with reason
- Success: 201 with the moderation action
- Errors: 404 when the service cannot be found, 409 for invalid transitions

GET /v1/admin/moderation/actions

- Auth: admin actor
- Request: service_id query parameter
- Success: 200 with the moderation action history
- Errors: 422 for invalid query values

Design Notes

- Quote creation is public. Consumers do not need to authenticate to request a quote.
- Discovery routes only expose public, active service data.
- Provider upstream URLs and credentials are never returned by discovery responses.
- The paid invoke flow is a two-step 402 challenge/settle flow backed by x402-compatible headers.
- The provider payout request path is separate from payout reporting and requires JWT auth plus idempotency protection.